

Ques

Evaluate $\int_{1/3}^3 \frac{1}{x} \ln \left(\left| \frac{x+x^2-1}{x-x^2+1} \right| \right) dx$

Soln

$$I = \int_{1/3}^3 \frac{1}{x} \ln \left| \frac{x+x^2-1}{x-x^2+1} \right| dx$$

put

$$x = \frac{1}{t}$$

$$dx = -\frac{dt}{t^2}$$

$$= \int_{1/3}^3 t \ln \left| \frac{\left(\frac{t+1-t^2}{t^2} \right)}{\left(\frac{t-1+t^2}{t^2} \right)} \right| \cdot \left(-\frac{dt}{t^2} \right)$$

$$= - \int_{1/3}^3 \frac{1}{t} \ln \left| \frac{t-1+t^2}{t+1-t^2} \right| dt = -I \Rightarrow \boxed{I=0} \quad \triangle$$

Que

Let $f(x)$ is continuous and positive function satisfying
 $\int_0^1 f(x) \cdot dx = 1$, $\int_0^1 x \cdot f(x) \cdot dx = 2$ and $\int_0^1 x^2 f(x) \cdot dx = 4$, find $f(x)$.

Sol

Given $f(x) > 0$

$$\int_0^1 f(x) \cdot dx = 1 \quad \text{--- (1)}$$

$$\int_0^1 x \cdot f(x) \cdot dx = 2 \quad \text{--- (2)}$$

$$\int_0^1 x^2 f(x) \cdot dx = 4 \quad \text{--- (3)}$$

Consider $4 \cdot (1) - 4 \cdot (2) + (3)$

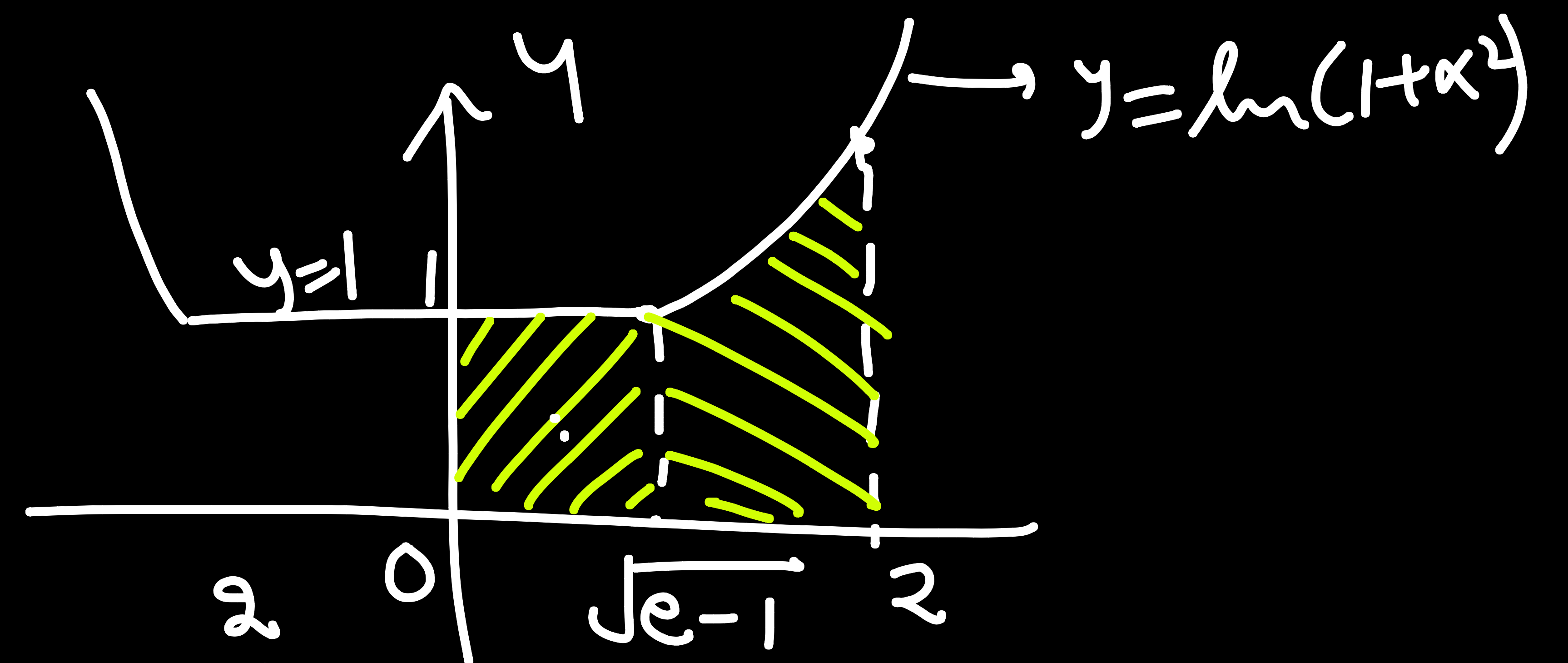
$$\int_0^1 f(x) (4 - 4x + x^2) \cdot dx = 4 \cdot 1 - 4 \cdot 2 + 4$$

$$\int_0^1 (x-2)^2 \cdot f(x) \cdot dx = 0$$

which is not possible as $(x-2)^2 \cdot f(x) > 0$
hence algebraic area
must have some positive value.

$\Rightarrow$ There is no such function $f(x)$.

Ques Evaluate $\int_0^2 \max. \{ \ln(1+x^2), 1 \} \cdot dx$



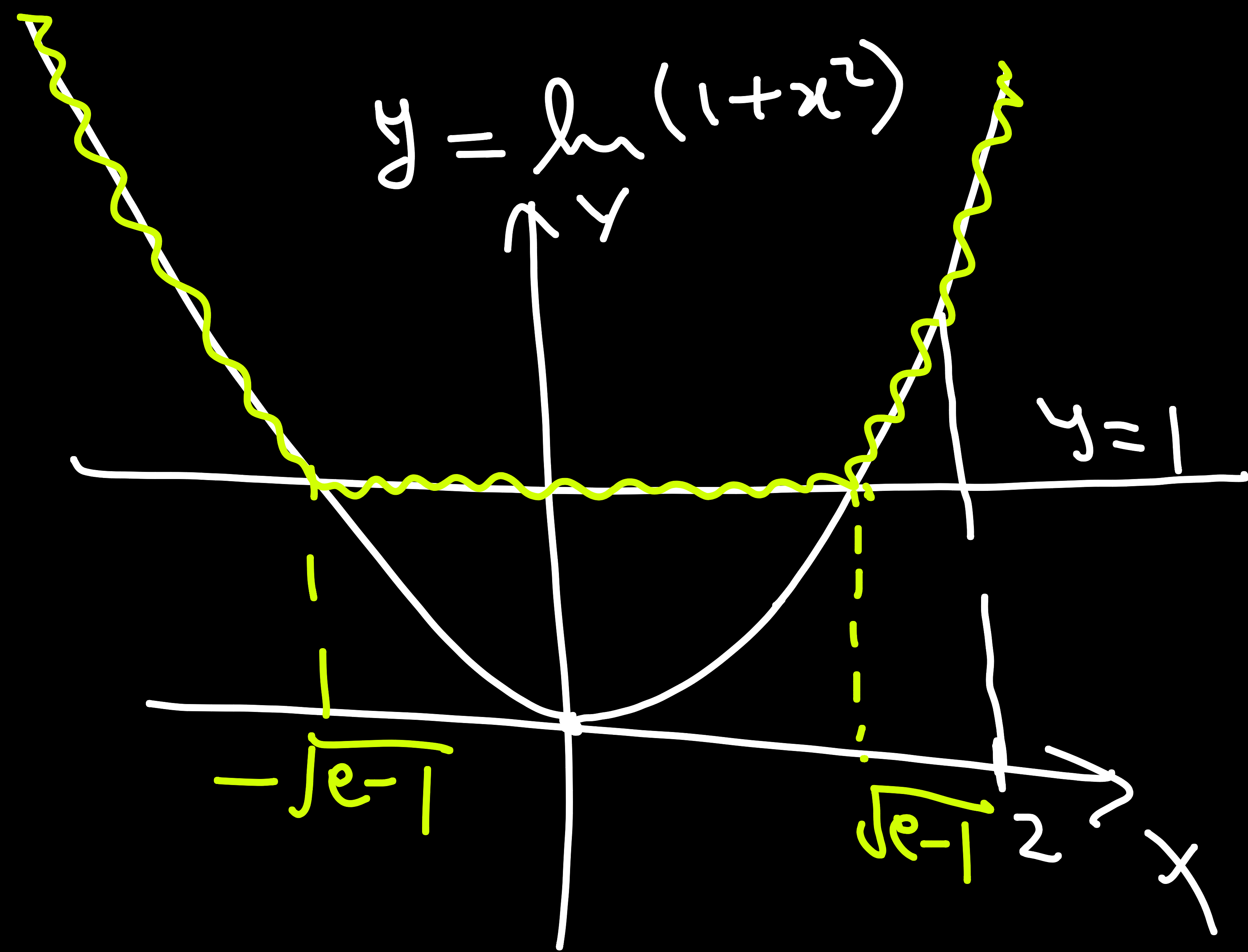
Sol $\int_0^2 \max. \{ \ln(1+x^2), 1 \} \cdot dx$

$$= 1 \cdot \sqrt{e-1} + \int_{\sqrt{e-1}}^2 \ln(1+x^2) \cdot dx$$

$$= \sqrt{e-1} + \left[x \cdot \ln(1+x^2) - \int \frac{2x^2}{1+x^2} \cdot dx \right]_{\sqrt{e-1}}^2$$

$$= \sqrt{e-1} + 2 \ln(5) - \sqrt{e-1} - 2 \int_{\sqrt{e-1}}^2 \frac{x^2+1-1}{x^2+1} dx$$

$$= 2 \ln(5) - 2 \left(x - \tan^{-1}(x) \right)_{\sqrt{e-1}}^2$$



$$\begin{aligned} \ln(1+x^2) &= 1 \\ 1+x^2 &= e \\ x^2 &= e-1 \\ x &= \pm \sqrt{e-1} \end{aligned}$$

$$= 2 \ln(5) - 2 \left(x - \tan^{-1}(x) \right)_{\sqrt{e-1}}^2$$

$$= 2 \ln(5) - 2 \left(2 - \tan^{-1}(2) - \sqrt{e-1} + \tan^{-1}(\sqrt{e-1}) \right)$$

A

Ques

Evaluate $\int_0^{\infty} \frac{x^4 + 1}{x^2 (x^2 + 1)^2} \cdot dx$

Sol $I = \int_0^{\infty} \frac{x^4 + 1}{x^2 (x^2 + 1)^2} \cdot dx$

$$x = \tan(\theta) \Rightarrow dx = \sec^2(\theta) \cdot d\theta$$

$$I = \int_0^{\pi/2} \frac{(\tan^4(\theta) + 1) \cdot \sec^2(\theta) \cdot d\theta}{\tan^2(\theta) \cdot \sec^4(\theta)}$$

$$= \int_0^{\pi/2} \left(\frac{s^4 + c^4}{s^2} \right) \cdot \frac{c^2 \cdot c^2}{s^2} d\theta = \int_0^{\pi/2} \left(\frac{s^4 + c^4}{s^2} \right) d\theta$$

$$\begin{aligned} &= \int_0^{\pi/2} \frac{(s^2 + c^2)^2 - 2s^2 c^2}{s^2} \cdot d\theta \\ &= \int_0^{\pi/2} \frac{1}{\sin^2(\theta)} d\theta - \int_0^{\pi/2} 2 \cos^2(\theta) \cdot d\theta \\ &= \left[-\cot(\theta) \right]_0^{\pi/2} - \text{Flunk} \\ &\quad \downarrow \text{D.N.E} \end{aligned}$$

$$= \underline{\underline{\text{D.N.E.}}}$$

Ques Evaluate $\int_1^{\infty} \frac{x^4+1}{x^2(x^2+1)^2} \cdot dx$

Sol

$$\int_1^{\infty} \frac{x^4 + 2x^2 + 1 - 2x^2}{x^2(x^2+1)^2} dx$$

$$= \int_1^{\infty} \frac{1}{x^2} dx - 2 \int_1^{\infty} \frac{dx}{(x^2+1)^2} \quad \left. \begin{array}{l} \\ \end{array} \right\} x = \tan(\theta)$$

$$= \left[-\frac{1}{x} \right]_1^{\infty} - 2 \int_{\pi/4}^{\pi/2} \frac{\sec^2(\theta) d\theta}{\sec^4(\theta)}$$

$$= 1 - \int_{\pi/4}^{\pi/2} (1 + \cos(2\theta)) d\theta$$

$$= 1 - \left[\theta + \frac{\sin(2\theta)}{2} \right]_{\pi/4}^{\pi/2} = 1 - \left(\frac{\pi}{4} + 0 - \frac{1}{2} \right) = \frac{3}{2} - \frac{\pi}{4} = \left(\frac{6-\pi}{4} \right)$$

Ques If $\int_0^\pi \theta^3 \ln(\sin \theta) d\theta = \lambda \frac{\pi}{4} \int_0^\pi \theta^2 \ln(\sqrt{2} \sin \theta) d\theta$, then find ' λ '.

$$\text{Sol } \text{L.H.S} = I = \int_0^\pi \theta^3 \ln(\sin \theta) d\theta = \int_0^\pi (\pi - \theta)^3 \ln(\sin \theta) d\theta$$

$$I = \pi^3 \int_0^\pi \ln(\sin \theta) d\theta - \underbrace{\int_0^\pi \theta^3 \ln(\sin \theta) d\theta}_{= I} + 3\pi \int_0^\pi \theta^2 \ln(\sin \theta) d\theta - 3\pi^2 \int_0^\pi \theta \ln(\sin \theta) d\theta$$

$$2I = \pi^3 (-\pi \ln(2)) + 3\pi \int_0^\pi \theta^2 \ln(\sin \theta) d\theta - 3\pi^2 I_1 \text{ (say)}$$

$$I_1 = \int_0^\pi \theta \ln(\sin \theta) d\theta = \int_0^\pi (\pi - \theta) \ln(\sin \theta) d\theta = \pi (-\pi \ln(2)) - I_1$$

$$I_1 = -\frac{\pi^2}{2} \ln(2) \left\{ \begin{aligned} 2I &= -\pi^4 \ln(2) + 3\pi \int_0^\pi \theta^2 \ln(\sin \theta) d\theta - 3\pi^2 \left(-\frac{\pi^2}{2} \ln(2)\right) \\ &= \frac{\pi^4}{2} \ln(2) + 3\pi \int_0^\pi \theta^2 \ln(\sin \theta) d\theta \end{aligned} \right.$$

$$I = \frac{\pi^4}{4} \ln(2) + \frac{3\pi}{2} \int_0^\pi \theta^2 \ln(\sin \theta) d\theta.$$

$$R.H.S. = \lambda \cdot \frac{\pi}{4} \int_0^\pi \theta^2 \ln(\sqrt{2} \sin \theta) \cdot d\theta$$

$$= \frac{\lambda \pi}{4} \left(\int_0^\pi \theta^2 \cdot \ln(\sqrt{2}) d\theta + \int_0^\pi \theta^2 \ln(\sin \theta) d\theta \right)$$

$$= \frac{\lambda \pi}{4} \cdot \frac{\pi^3}{6} \cdot \ln(2) + \frac{\lambda \pi}{4} \int_0^\pi \theta^2 \ln(\sin \theta) d\theta$$

Comparing with I we will get $\boxed{\lambda = 6}$

Que

Evaluate $\int_{-\frac{1}{\sqrt{3}}}^{\frac{1}{\sqrt{3}}} \frac{\cos^{-1}\left(\frac{2x}{1+x^2}\right) + \tan^{-1}\left(\frac{2x}{1-x^2}\right)}{e^x + 1} \cdot dx$

Sol $x = \tan \theta \Rightarrow \theta \in \left[-\frac{\pi}{6}, \frac{\pi}{6}\right]$
 $\hookrightarrow 2\theta \in \left[-\frac{\pi}{3}, \frac{\pi}{3}\right]$

$$\begin{aligned} \cos^{-1}\left(\frac{2x}{1+x^2}\right) &= \cos^{-1}\left(\frac{2\tan\theta}{1+\tan^2\theta}\right) \\ &= \cos^{-1}(\sin(2\theta)) \\ &= \frac{\pi}{2} - \sin^{-1}(\sin(2\theta)) \\ &= \frac{\pi}{2} - 2\theta \\ \tan^{-1}\left(\frac{2x}{1-x^2}\right) &= \tan^{-1}(\tan(2\theta)) = 2\theta \end{aligned}$$

$\uparrow \frac{\pi}{2}$

$$= \int_{-\frac{1}{\sqrt{3}}}^{\frac{1}{\sqrt{3}}} \frac{\pi/2}{e^x + 1} \cdot dx$$

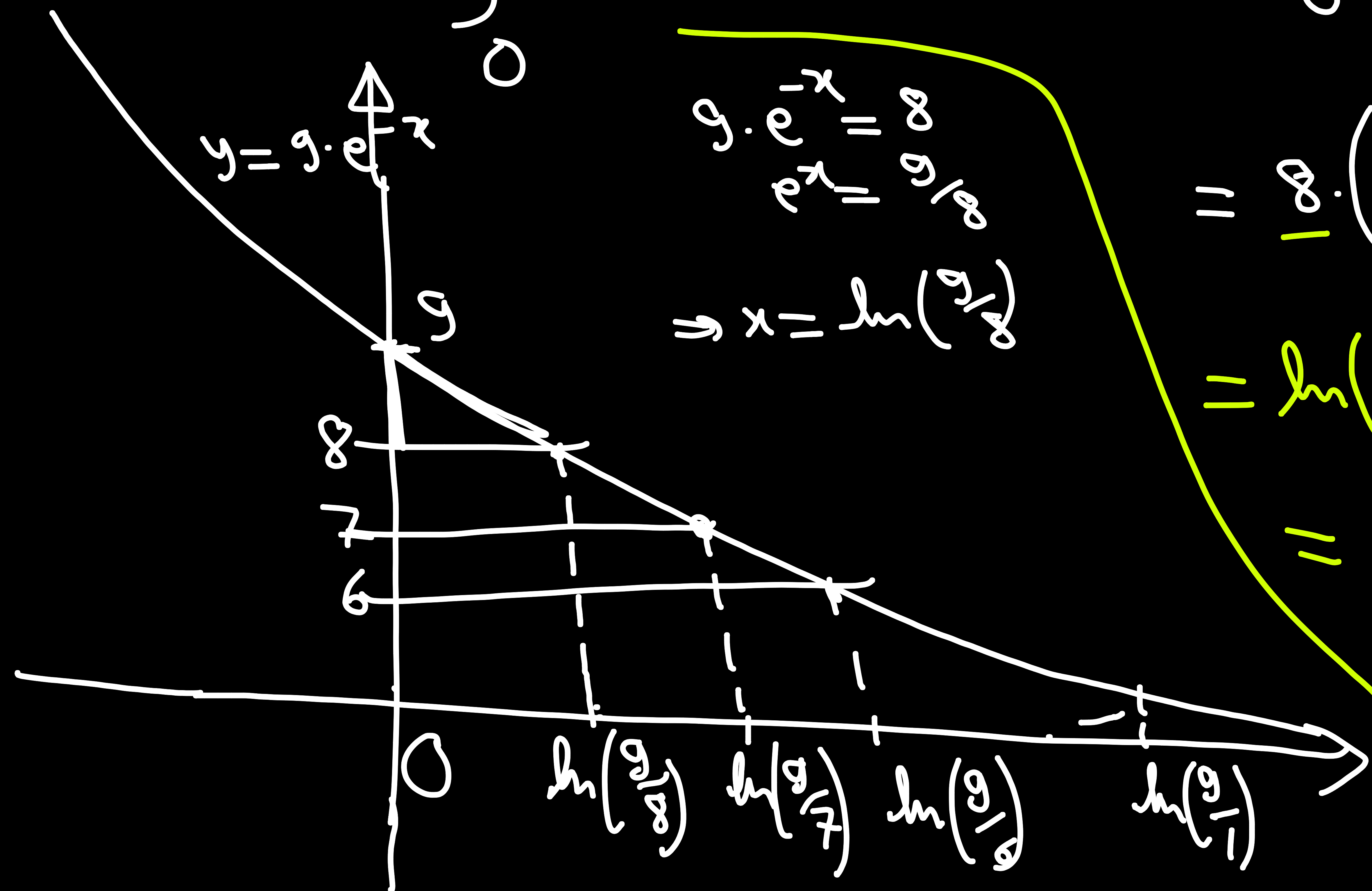
$$\begin{aligned} &= \frac{\pi}{2} \int_{-\frac{1}{\sqrt{3}}}^{\frac{1}{\sqrt{3}}} \left(\frac{1}{e^x + 1} + \frac{1}{e^{-x} + 1} \right) dx \\ &= \frac{\pi}{2} \int_0^{\frac{1}{\sqrt{3}}} \frac{1+e^x}{1+e^x} dx = \frac{\pi}{2} \cdot \frac{1}{\sqrt{3}} = \frac{\pi}{2\sqrt{3}} \end{aligned}$$

Ques

Evaluate $\int_0^{\infty} \left[\frac{g}{e^x} \right] \cdot dx$, where $[.]$ denotes G.I.F.

Sol

$$\int_0^{\infty} [g e^{-x}] \cdot dx = \int_0^{\ln(\frac{g}{8})} 8 \cdot dx + \int_{\ln(\frac{g}{8})}^{\ln(\frac{g}{7})} 7 \cdot dx + \int_{\ln(\frac{g}{7})}^{\ln(\frac{g}{6})} 6 \cdot dx + \dots + \int_{\ln(\frac{g}{2})}^{\ln(\frac{g}{1})} 1 \cdot dx + \int_{\ln(5)}^{\infty} 0 \cdot dx$$



$$g \cdot e^{-x} = 8$$

$$e^x = \frac{g}{8}$$

$$\Rightarrow x = \ln\left(\frac{g}{8}\right)$$

$$= 8 \cdot \left(\ln\left(\frac{g}{8}\right) - 0 \right) + 7 \cdot \left(\ln\left(\frac{g}{7}\right) - \ln\left(\frac{g}{8}\right) \right) + 6 \cdot \left(\ln\left(\frac{g}{6}\right) - \ln\left(\frac{g}{7}\right) \right) + \dots + 1 \cdot \left(\ln(g) - \ln\left(\frac{g}{2}\right) \right)$$

$$= \ln\left(\frac{g}{8}\right) + \ln\left(\frac{g}{7}\right) + \ln\left(\frac{g}{6}\right) + \dots + \ln\left(\frac{g}{2}\right) + \ln(g)$$

$$= \ln\left(\frac{g^8}{8}\right) = \ln\left(\frac{g^9}{19}\right)$$

Ans

Ques

gf $\int_0^{\infty} \frac{dx}{1+x^{2021}} = \int_0^1 \frac{dx}{(1-x^{2021})^{\lambda}}$, then find ' λ '.

Sol

$$\text{L.H.S.} = \int_0^{\infty} \frac{dx}{1+x^{2021}} = \int_0^{\infty} \frac{dx}{1+\left(x^{\frac{2021}{2}}\right)^2}$$

$$= \frac{2}{2021} \int_0^{\pi/2} \frac{\tan^{-\frac{2019}{2021}}(\theta) \cdot \cancel{\sec^2(\theta)} d\theta}{\cancel{\sec^2(\theta)}}$$

$$= \frac{2}{2021} \int_0^{\pi/2} \tan^{-\frac{2019}{2021}}(\theta) \cdot d\theta.$$

$$x^{\frac{2021}{2}} = \tan(\theta)$$

$$x = \tan^{\frac{2}{2021}}(\theta)$$

$$dx = \frac{2}{2021} \tan^{\frac{2}{2021}-1}(\theta) \cdot \sec^2(\theta) d\theta$$

$$R.H.S. = \int_0^1 \frac{dx}{(1-x^{2021})^\lambda}$$

$$= \frac{2}{2021} \int_0^{\pi/2} \frac{x^{-\frac{2019}{2021}} \cdot c}{(c^2)^\lambda} \cdot d\theta$$

$$= \frac{2}{2021} \int_0^{\pi/2} \frac{x^{-\frac{2019}{2021}}}{c^{2\lambda-1}} \cdot d\theta$$

$$L.H.S. = \frac{2}{2021} \int_0^{\pi/2} \tan^{\frac{-2019}{2021}}(\theta) d\theta = \frac{2}{2021} \int_0^{\pi/2} \frac{\sin^{\frac{-2019}{2021}}(\theta)}{\cos^{\frac{-2019}{2021}}(\theta)} d\theta$$

$$x^{\frac{2021}{2}} = \sin(\theta)$$

$$x = \sin^{\frac{2}{2021}}(\theta)$$

$$dx = \frac{2}{2021} \sin^{\frac{2}{2021}-1}(\theta) \cdot \cos(\theta) d\theta$$

Compare

$$2\lambda - 1 = -\frac{2019}{2021}$$

$$2\lambda = \frac{2}{2021}$$

$$\lambda = \frac{1}{2021}$$

✓

Ques If $I(u) = \int_0^\pi \ln(1 - 2u \cos(x) + u^2) \cdot dx$, then find $\frac{I(u) + I(-u)}{I(u^2)}$.

Sol

$$I(u) = \int_0^\pi \ln(1 - 2u \cos(x) + u^2) \cdot dx$$

$$I(-u) = \int_0^\pi \ln(1 + 2u \cos(x) + u^2) \cdot dx = \int_0^\pi \ln(1 - 2u \cos(x) + u^2) \cdot dx = I(u).$$

$\xrightarrow{a+b-x}$
 $\hookrightarrow x \rightarrow \pi - x$

$\downarrow$

$$I(u) + I(-u) = \int_0^\pi \ln((1 + 2u \cos(x) + u^2)(1 - 2u \cos(x) + u^2)) \cdot dx$$

$$= \int_0^\pi \ln((1+u)^2 - 4u^2 \cos^2(x)) \cdot dx = \int_0^\pi \ln(1 + u^4 + 2u^2 - 4u^2 \cos^2(x)) \cdot dx$$

$\hookrightarrow 2u^2(1 - 2\cos^2(x))$
 $= 2u^2(-\cos(2x))$

$$\begin{aligned}
 I(u) + I(-u) &= \int_0^{\pi} \ln(1 + u^4 - 2u^2 \cos(2x)) \cdot dx \\
 &\quad \downarrow \underline{0-2\pi} \\
 &= 2 \int_0^{\pi/2} \ln(1 + u^4 - 2u^2 \cos(2x)) \cdot dx \\
 &\quad \begin{array}{l} 2x = t \\ dx = \frac{dt}{2} \end{array} \\
 &= 2 \int_0^{\pi} \ln(1 + u^4 - 2u^2 \cos(t)) \cdot \frac{dt}{2} = \int_0^{\pi} \ln(1 + u^4 - 2u^2 \cos(t)) dt \\
 &= I(u^2)
 \end{aligned}$$

$$\Rightarrow I(u) + I(-u) = I(u^2)$$

$$\Rightarrow \frac{I(u) + I(-u)}{I(u^2)} = \boxed{1} \quad \text{Q.E.D.}$$

Ques

Evaluate $\int_0^{\infty} \frac{dx}{1+x^4}$.

Sol

$$I = \int_0^{\infty} \frac{dx}{1+x^4}$$

$$\begin{aligned} x &= \frac{1}{t} \\ dx &= -\frac{dt}{t^2} \end{aligned} \quad \int_{\infty}^0 \frac{-\frac{dt}{t^2}}{\left(\frac{1+t^4}{t^4}\right)}$$

$$I = \int_0^{\infty} \frac{t^2 dt}{1+t^4} = \int_0^{\infty} \frac{x^2 dx}{1+x^4}$$

$$+ \quad 2I = \int_0^{\infty} \frac{1+x^2}{1+x^4} dx = \int_0^{\infty} \frac{1+\frac{1}{x^2}}{x^2+\frac{1}{x^2}} dx$$

$$= \int_0^{\infty} \frac{1+\frac{1}{x^2}}{\left(x-\frac{1}{x}\right)^2+2} dx = \frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{x-\frac{1}{x}}{\sqrt{2}} \right) \Bigg|_0^{\infty} = \frac{1}{\sqrt{2}} \left(\frac{\pi}{2} - \left(-\frac{\pi}{2} \right) \right) = \frac{\pi}{\sqrt{2}}$$

$$2I = \frac{\pi}{\sqrt{2}}$$

$$\Rightarrow I = \frac{\pi}{2\sqrt{2}}$$

Ques Evaluate $\int_0^1 \cot^{-1}(1-x+x^2) \cdot dx$

Sol $I = \int_0^1 \cot^{-1}(1-x+x^2) \cdot dx$

$$= \int_0^1 \tan^{-1}\left(\frac{1}{1-x+x^2}\right) \cdot dx$$

$$= \int_0^1 \tan^{-1}\left(\frac{(1-x)+x}{1-x \cdot (1-x)}\right) dx$$

$$= \int_0^1 (\tan^{-1}(1-x) + \tan^{-1}(x)) dx$$

$$= \int_0^1 \tan^{-1}(1-x) dx + \int_0^1 \tan^{-1}(x) dx = 2 \int_0^1 \tan^{-1}(x) dx$$

$$I = 2 \int_0^1 \tan^{-1}(x) dx$$

$$= 2 \left(\left[x \cdot \tan^{-1}(x) \right]_0^1 - \int_0^1 \frac{x}{1+x^2} dx \right)$$

$$= 2 \left(1 \cdot \frac{\pi}{4} - 0 \right) - \left[\ln(1+x^2) \right]_0^1$$

$$= \frac{\pi}{2} - \ln(2) \quad \underline{\underline{\Delta}}$$

Ques

Evaluate $\int_0^1 x^m (1-x)^n \cdot dx$, where $m, n \in \mathbb{N}$.

Sol $I = \int_0^1 x^m (1-x)^n \cdot dx \xrightarrow{m-1} \int_0^{\pi/2} \sin^{2m}(\theta) \cdot (c^2)^n \cdot 2 \sin \theta \cdot d\theta$
 $\Rightarrow$ Wallis's Theorem

m-2 Consider $(1-x+t \cdot x)^{m+n} = \sum_{r=0}^{m+n} C_r \cdot (1-x)^{m+n-r} (tx)^r$
 $\int_0^1 (1+x(t-1))^{m+n} \cdot dx = \sum_{r=0}^{m+n} C_r \cdot t^r \cdot \int_0^1 (1-x)^{m+n-r} \cdot x^r \cdot dx$

$\int_0^1 \frac{(1+x(t-1))^{m+n+1}}{(m+n+1) \cdot (t-1)} = \left[\frac{t^{m+n+1} - 1}{(t-1)(m+n+1)} \right] = \frac{1}{m+n+1} \left(1+t+t^2+\dots+t^{m+n} \right)$

$$\frac{1}{m+n+1} (1+t+t^2+\dots+t^{m+n}) = \sum_{r=0}^{m+n} {}^{m+n}C_r \cdot t^r \cdot \int_0^1 (1-x)^{m+n-r} \cdot x^r \cdot dx$$

$$= \left(\int_0^1 (1-x)^{m+n} \cdot dx \right) + t \left(\int_0^1 (1-x)^{m+n-1} \cdot x \cdot dx \right) + t^2 \left(\int_0^1 (1-x)^{m+n-2} \cdot x^2 \cdot dx \right)$$

$$+ \dots + t^m \left({}^{m+n}C_m \cdot \int_0^1 (1-x)^n \cdot x^m \cdot dx \right) + \dots$$

Comparing coeff of t^m on both sides

$$\frac{1}{m+n+1} = {}^{m+n}C_m \cdot \int_0^1 (1-x)^n \cdot x^m \cdot dx$$

$$\int_0^1 (1-x)^n \cdot x^m \cdot dx = \frac{1}{({}^{m+n}C_m \cdot (m+n+1))} = \boxed{\frac{{}^n C_m}{m+n+1}}$$

Ques If $\int_0^{\infty} f\left(\frac{a}{x} + \frac{x}{a}\right) \cdot \frac{\ln(x)}{x} \cdot dx = \lambda \cdot \int_0^{\infty} f\left(\frac{a}{x} + \frac{x}{a}\right) \cdot \frac{dx}{x}$ then find λ .
 given $a > 0$ is positive real constant.

Sol L.H.S = $I = \int_0^{\infty} f\left(\frac{a}{x} + \frac{x}{a}\right) \cdot \frac{\ln(x)}{x} \cdot dx$

$$\frac{x}{a} = \frac{a}{t}$$

$$dx = -\frac{a^2}{t^2} \cdot dt$$

$$= \int_{\infty}^0 f\left(\frac{t}{a} + \frac{a}{t}\right) \cdot \frac{\ln\left(\frac{a^2}{t}\right)}{\left(\frac{a^2}{t}\right)} \cdot \left(-\frac{a^2}{t^2} \cdot dt\right)$$

$$= \int_0^{\infty} f\left(\frac{t}{a} + \frac{a}{t}\right) \cdot \frac{\ln\left(\frac{a^2}{t}\right)}{t} \cdot dt = \int_0^{\infty} f\left(\frac{t}{a} + \frac{a}{t}\right) \cdot \frac{\ln(a^2)}{t} \cdot dt - \int_0^{\infty} f\left(\frac{t}{a} + \frac{a}{t}\right) \cdot \frac{\ln(t)}{t} \cdot dt$$

$$\cancel{I} = \cancel{\ln(a)} \int_0^{\infty} f\left(\frac{t}{a} + \frac{a}{t}\right) \cdot \frac{dt}{t} \Rightarrow I = \ln(a) \int_0^{\infty} f\left(\frac{t}{a} + \frac{a}{t}\right) \cdot \frac{dt}{t} \quad \underbrace{\qquad\qquad\qquad}_{= I}$$

Comparing

$$\lambda = \frac{\ln(a)}{A}$$

A